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Event triggers and opinion leaders shape climate change discourse on Weibo

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Understanding how real-world events and opinion leaders shape climate change discussions is vital for improving communication and policy formulation to meet global carbon mitigation goals. This study analyzed 5.3 million original posts from Weibo (2012–2022), China's largest social media platform, to examine climate change discourse. We found five event types triggering 48 discussion peaks, including online activities, international conferences, extreme weather, domestic policies, and international news. Posts generally conveyed positive attitudes, though sentiment decreased during haze pollution and the COVID-19 pandemic. Network analysis revealed seven opinion leader groups with distinct strategies: official media and institutions emphasized political will, global initiatives, and socio-economic implications, while universities and grassroots individuals focused on scientific reality and personal actions. Celebrities and unofficial accounts often highlighted geopolitical topics, especially China-US relations. We suggest reducing fragmented echo chambers and fostering personal connections through digital media platforms to enhance public awareness.

Climate change is one of the most pressing challenges confronting humanity today. Despite climate actions being primarily led by governments, public opinions play an increasingly significant role in shaping the formulation, implementation, and effectiveness of policies[1]. Climate change has transcended the realm of scientific debate and evolved into a multifaceted issue that influences not only environmental policies but also social[2], political[3], and livelihood discussions[4]. For example, climate change has emerged as a good reflection of political polarization and population segregation, with divergent views aligning along ideology and partisanship[5]. In this context, comprehending public discourse of climate change is instrumental in designing more targeted climate policies[6]. It also sheds light on the influence of social culture and media messaging on public viewpoints[7,8].

As the world's largest emitter, China is playing a crucial role in global mitigation and adaptation effort. Yet, research on public discourse of climate change in China remains limited, often relying on surveys that assess awareness or policy support[9–11]. While valuable, these studies frequently lack the timeliness and depth needed to capture how the public interprets climate issues and engages in discussions spurred by specific real-world events[12]. With the rapid growth of digital industries, a wide range of social media platforms has emerged, offering opportunities to publicize environmental concerns, foster knowledge-sharing, and support governance[13–15]. Over the last decades, China's climate communication models have also evolved, with increasing engagement on online platforms[16]. However, the key opinion leaders, their networks, and framing strategies are largely uncharted, leaving a gap in understanding how public discourse is shaped as a result[9]. To develop effective and tailored strategies for future climate communication, it is essential to reveal underlying mechanisms that drive discourse spread.

Social media provides valuable benefits by offering more precise temporal granularity and a detailed network of connections that facilitate the spread of discourse[17]. However, feed algorithms may narrow the range of content users see, primarily suggesting posts similar to those they already engage with, which can introduce challenges[18]. For instance, social media can restrict exposure to varied perspectives, fostering the formation of like-minded groups and reinforcing shared narratives, an effect known as echo chambers[19]. According to group polarization and social identity theories, the framing and agenda-setting within these echo chambers prompt individuals to seek information that aligns with their beliefs and social identity, deepening existing biases[20]. These effects can undermine climate action by polarizing public views and fostering the spread of misinformation[21].

In China, climate change discourse has traditionally followed a top-down model, dominated by government agencies and news media[9]. Framing analyses of mainstream media and newspapers reveal a focus on large-scale narratives[22], such as geopolitical boundaries of achievements and attribution of responsibility. However, as communication increasingly shifts to digital social media platforms, a broader array of opinion leaders now engages in climate communication in China, such as non-governmental

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organizations, celebrities, and grassroots groups. Recent studies suggest that both the primary topics and the actors in climate change discussions on Chinese social media differ from those in traditional media[16]. Nevertheless, a significant theoretical gap remains regarding how official and non-official actors interact in these discussions. Given that audiences may have varying preferences for information sources, it is unclear whether echo chambers have formed among these actors, potentially hindering collective consensus and action on climate issues.

In this study, we conduct a comprehensive descriptive data mining analysis of Sina Weibo, one of China's largest social media platforms, to examine how climate change discourses are shaped by various actors and their framing strategies (Fig. S1). Sina Weibo, launched in 2009, has emerged as the country's foremost social media platform, having 260 million daily active users as of the first quarter of 2023[23]. Similar to Twitter (now 'X'), Weibo allows people to voice and exchange opinions, with influencers actively engaging in discussions. Despite criticism for potential robot accounts on the Weibo platform, overall, given its vast user base in China, Weibo remains an ideal dataset for exploring public discourse on climate change[24–26].

To investigate how official and non-official actors interact and participate in climate change discussions on Weibo, we develop an analytical framework that addresses key research questions regarding the triggering events, opinion leader networks, and framing strategies surrounding climate change topics. RQ1: How has public attention to climate change evolved over time, and what real-world events have triggered these discussions? RQ2: Who are the main opinion leaders driving the discourse, and how does opinion spread and interact among users? RQ3: What are the predominant frames of climate change frequently used by opinion leaders?

To address these questions, this study analyzed 5.3 million original posts containing climate change keywords on Weibo, spanning from 2012 to 2022. We first developed an Online Attention Metric (OAM) index to track weekly fluctuations in the intensity of online debates. To link these peaks with specific real-world events, we applied the Biterm Topic Model (BTM) to summarize the main topics in the Weibo posts associated with each peak. This topic modeling technique captures co-occurrence patterns of word pairs (i.e., biterms) across the entire corpus, making it particularly effective for analyzing short texts, such as social media posts or forum discussions. We identified 48 real-world events that triggered spikes in discussion (Table S1). Using network analysis, we identified seven groups of opinion leaders who shape the discourse and mapped their interaction networks. Finally, we applied text analysis and language models to parse the frames frequently adopted by different actors during specific real-world events.

Based on the identification of opinion leaders, we categorized them as having either official or unofficial backgrounds by reviewing their verified status and personal profiles. Drawing on group polarization and social identity theories[27,28], audiences cluster around different types of opinion leaders, reinforcing their social identities through establishing relatively fixed frames and viewpoints within their respective echo chambers. Therefore, we propose the following hypothesis:

H1: Opinion leaders from official and unofficial backgrounds create echo chambers, with each group engaging in unique ways in online discussions triggered by real-world climate events.

In addition, according to the message framing and agenda-setting theories[29,30], media and opinion leaders may influence audience perceptions by emphasizing certain topics and presenting information in specific ways, shaping the public understanding and perceived importance of climate issues. Thus, we further hypothesize:

H2: The framing of climate change messages varies with specific events, and the prevalence of certain frames differs between official and unofficial opinion leaders.

Results

Event-driven climate change online discussions in China

Using our self-developed Online Attention Metric (OAM) index, we tracked the intensity and engagement of climate change discussions on Weibo, revealing a gradual upward trend with periodic spikes (Fig. S2). After March 2018, there was a marked surge in attention toward climate change, nearly doubling in volume (Fig. S3). The fluctuating trend indicates that climate change discussions on the Chinese social media platform are strongly driven by specific events.

We then used the z-score method to detect all instances of peak OAM data from 2012 to 2022, identifying a total of 48 significant peaks. To link these peaks with specific real-world events, we applied the Biterm Topic Model (BTM) to summarize the main topics in the Weibo posts associated with each peak. This topic modeling technique captures co-occurrence patterns of word pairs (i.e., biterms) across the entire corpus, making it particularly effective for analyzing short texts, such as social media posts or forum discussions[31]. Based on the topics extracted by BTM, we manually verified the associated real-world events and categorized them into five groups: Online Activities, International Conferences, Extreme Weather, Domestic Policy, and International News (Fig. S4).

Figure 1 illustrates specific events within each group alongside their posting timelines. A general trend observed is the increased prominence of discussions on domestic policies and international diplomacy news related to climate change after 2018. Online Activities contain 12 events (e.g., the Earth Hour) about online environmental movements aimed to raise awareness about climate change. International Conferences refer to global climate change gatherings such as the United Nations Climate Change Conference of the Parties (COPs). Extreme Weather includes severe weather phenomena associated with climate change, with summer heatwave garnering the most attention. Domestic Policy refers to policy trends, conferences, and news regarding domestic affairs. International News is mainly about China's participation in global climate change governance, or the dynamic of China-US diplomatic relations.

We further profiled users who participated in climate change discussions. Based on verification status on Weibo, we observed that a high proportion of blue-V users (enterprises, institutions, and media organizations) shared posts related to International Conferences, while a greater number of yellow-V users (mainly celebrities or self-media creators) were involved in posts of International News (Fig. S5). General unverified accounts were found to be more active in expressing opinions regarding Domestic Policies. Based on the account followers (Fig. S6), we noted substantial variations in the levels of influence among participants in each topic. International Conferences tended to feature highly influential figures (with over 1 million followers). Conversely, topics related to the Domestic Policies attracted followers numbering in the few hundred range.

Finally, we located users who posted original weibos during the peak events and reported their geographical addresses (Fig. S7). The majority of users were located in the highly developed Eastern Coastal provinces, notably Beijing, Shanghai, and Guangdong. A substantial number of Weibo posts originated from users with IP addresses in foreign countries. Accounts from the United States ranked within the top eight, alongside all Chinese provinces, likely influenced by the presence of global institutions like the United Nations headquartered there. These findings highlight the active engagement of international organizations on Chinese social media platforms.

To provide deeper insights into public attitudes toward these issues, we calculated sentiment scores for each Weibo post during the 48 peak events, with scores ranging from 0 to 100. Figure 2 shows the annual sentiment score results, highlighting the temporal trend over time. We also conducted qualitative content analysis to explore the themes in user-generated text (Table S2).

We found during the winter of 2013, a notable decrease in emotional expression was evident, primarily attributed to the haze crisis in January 2013. We observed low median sentiment scores of 47, 51, and 51 in response to event #8 (Haze event), event #5 (Haze control policy), and event #6 (Earth Hour). We also noticed a general decreasing trend in public emotions following the global COVID-19 pandemic in 2020. Apart from events related to domestic policies where local mainstream voices may dominate, the average sentiment for all other events has generally decreased

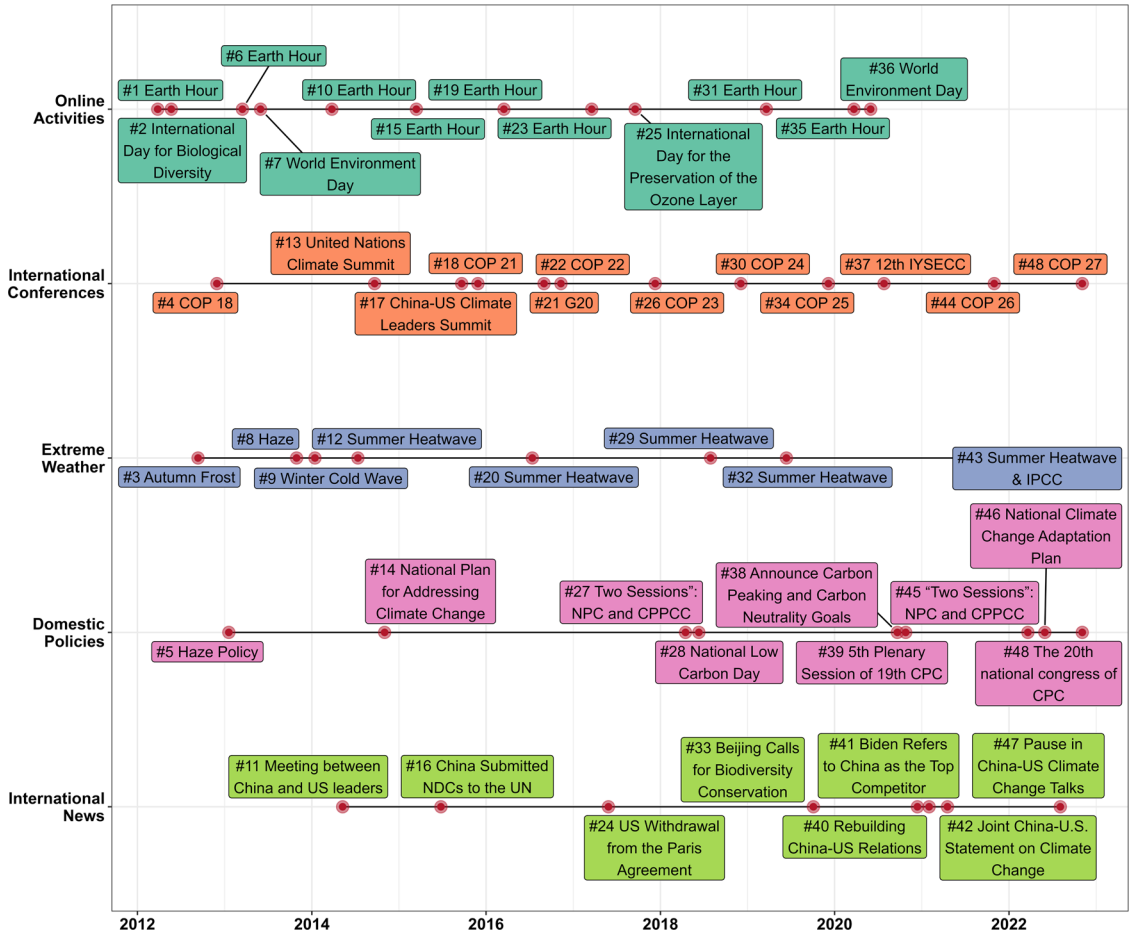

Fig. 1 | The timeline of 48 real-world peak events. The five primary event categories were delineated using different colors. Each event is briefly described with a short summary of its main content.

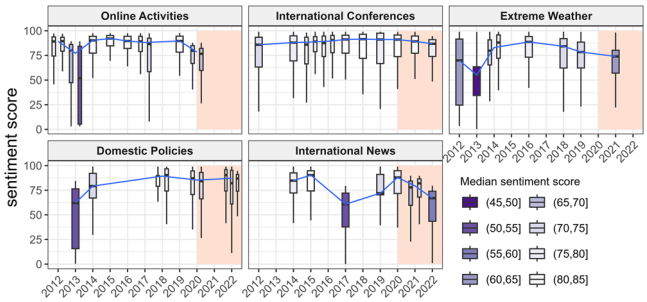

Fig. 2 | The sentiment scores of climate change discussions on Weibo during 48 peak events. Boxplots depict the distribution of sentiment scores per event, where higher score denotes positive emotion and lower score denotes negative emotion. Each box is color-coded based on the median value, with darker shades representing lower scores. Light red rectangles highlight events occurring after 2020, the global COVID-19 pandemic. Blue lines illustrate the temporal trend of annual median sentiments.

since 2020. The t-test conducted on the two sets of data before and after COVID-19 showed a significant change in emotions (Table S3)[32,33].

Despite the influence of the 2013 haze crisis and the COVID-19 pandemic, widespread public engagement with online activities and international conferences generally evoked positive emotions, with average scores exceeding 80. Discussions about domestic climate policies have also intensified, particularly in recent years, and have elicited positive emotions.

Public response to international news exhibits more fluctuations, especially concerning news involving US-China relations. The two lowest peak sentiment scores have median of 52 and 56 for International News following the U.S. announcement of its withdrawal from the Paris Agreement (event #5), and when climate talks between China and the U.S. were temporarily stalled (event #48).

Networks of online climate change opinion leaders

To identify climate change opinion leaders on China's social media platform and analyze their interactions, we developed a social network model based on text similarity. This approach addresses the challenge that Weibo data collected via APIs lacks post-repost relationships between accounts. It also has the advantage of identifying influencers who share similar opinions, even if they have no direct interactions. High text similarity between two users indicates shared attitudes toward climate change, placing them closer in the network with stronger connections. For each event category, we selected the event with the highest OAM score as a representative to construct the networks, as illustrated in Fig. 3. To enhance clarity, we included only the top 100 most influential users per event, measured by the combined total of comments, likes, and reposts on their posts. Table S4 shows that the number of followers and official background are the main factors for users to become opinion leaders.

The accounts of opinion leaders during these real-world events were manually reviewed and were categorized into three groups based on their account names, self-descriptions, and primary content, which included news media, institutions, and personal and business accounts (Fig. S8). Within the news media category, three distinct types of outlets were identified: central official media, local media and unofficial media,

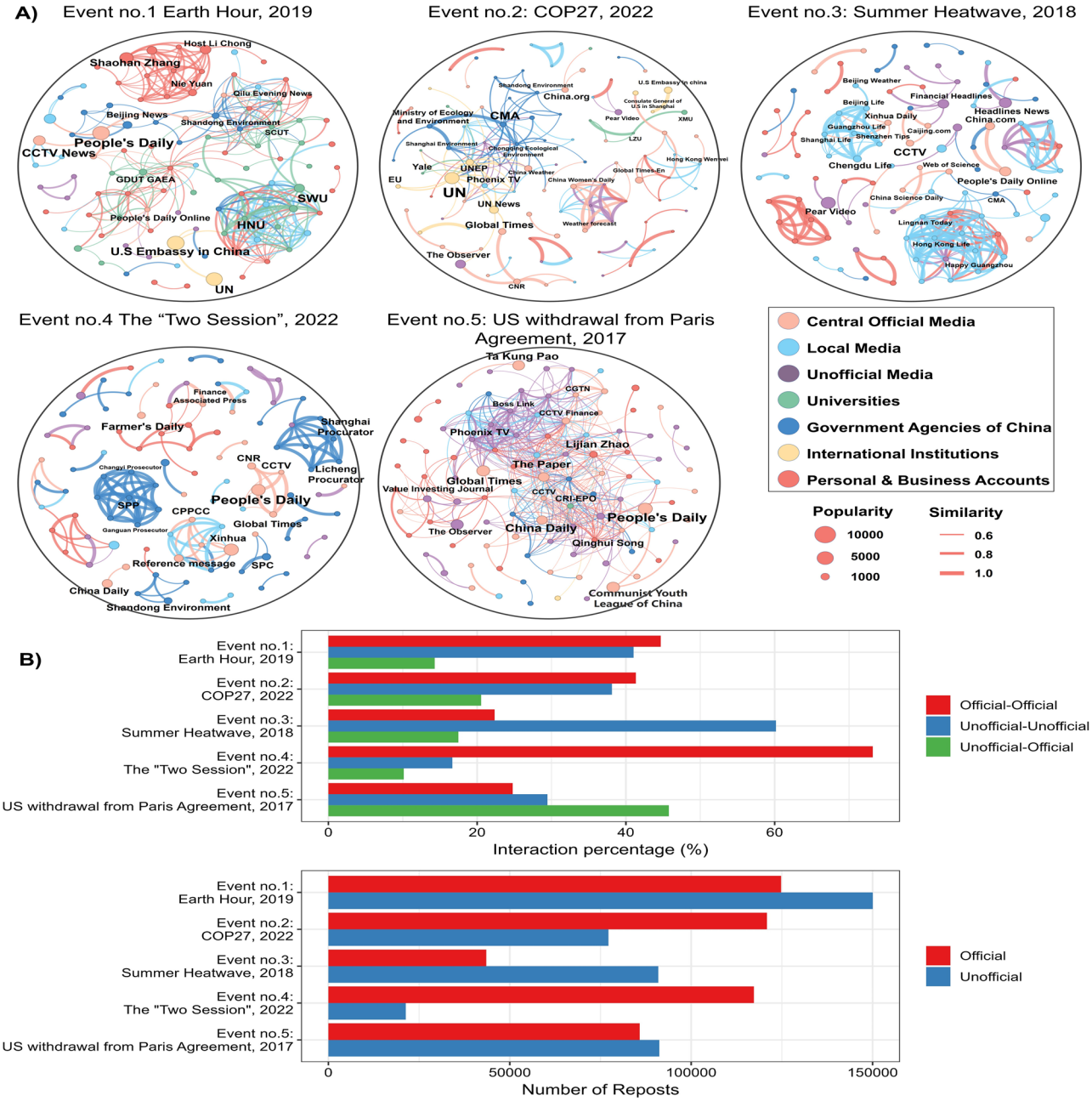

Fig. 3 | Opinion leaders and their social networks identified by text similarity in five representative peak events. A Seven distinct types of opinion leaders were identified and distinguished using different colors. Their social networks, based on text similarities in Weibo posts, were visualized in a circular layout for the five representative peak events. The size of each node corresponds to its influence, measured by the total number of reposts. The usernames of top opinion leaders are indicated on the plot. The transparency of edges reflects the text similarity between the connected users. **B** The upper panel illustrates the percentage of connections within and between official and unofficial user nodes. The lower panel depicts the influence of each opinion leader group across different events, measured by the total number of reposts.

categorized based on their ownership structures. Institutions were classified into three types, including universities, government agencies of China, and international institutions. Personal accounts mainly comprised enterprises, celebrities, and grassroots individuals. Table S5 lists the top 20 opinion leaders' names and their account types for each representative event.

These opinion leaders exhibited varying activities observed during each peak event. In analyzing the "Online Activities" event group, we focused on the Earth Hour initiative in 2019. We found that among the top 100 influential accounts, personal accounts (40) and universities (20) emerged as the most actively engaged entities. They formed two distinct clusters with tight connections. Local media (13) and government agency accounts (12) also showed notable activity but with loose connections. Central official media and international institutions are fewer in number and relatively isolated but yield significant influence in public engagement.

For the "International Conferences," we chose the COP27 event held in 2022 as the case. Our analysis revealed the presence of a single large cluster, with central nodes predominantly occupied by international institutions with the highest influence. Key players included prominent entities such as the United Nations (UN) and the UN Environment Programme (UNEP). Moreover, we observed a growing significance attributed to central official media (25), local media (17), and government agencies (15) within this cluster.

The network dynamics observed during the 2018 summer heatwave event were notably diverse, characterized by active involvement of local media (37) and personal & business accounts (25). We noted only a few

small networks, a pattern likely influenced by geographical disparities. Users situated in various locations may engage with the event in distinct manners.

During the "Two Sessions" in 2022, one of the most important political gatherings in China, we found central official agencies (20), such as People's Daily, CCTV, and Xinhua News Agency, produced highly influential weibo posts in addressing climate change. Several small networks were formulated by these accounts. Subsequently, a high level of participation was observed from accounts belonging to government agencies of China (40) (Fig. S9). These accounts played a crucial role in interpreting policies.

Regarding "International News," we chose the event of Trump's announcement to withdraw from the Paris Agreement in 2017. We observed the largest network with the highest number of connections (Fig. S10). Central official media (25), unofficial media (32), and personal and business accounts (24) collectively lead online discussion. Key nodes continue to be dominated by central official media outlets. For unofficial media, prominent accounts are mostly current affairs commentary platforms with large user base, such as Observer and Phoenix News. Personal accounts of self-media creators and individual commentators also actively engage in the discussion.

To test Hypothesis #1, we conducted two quantitative analyses to determine whether official and unofficial users have formed two independent echo chambers. We categorized the seven types of opinion leaders into either official or unofficial groups based on their profiles. The upper panel of Fig. 3B illustrates the interactions within and between these two groups, represented as percentages of all node connections. The results indicate that interactions within each group exceed those across groups, particularly for discussions about the Earth Hour and COP27. For topics such as the summer heatwave and the Two Sessions political gathering, conversations are dominantly driven by unofficial and official accounts respectively. Interestingly, the U.S. withdrawal from the Paris Agreement stands out, as it generates significantly higher interconnections between official and unofficial users than within either group. The lower panel of Fig. 3B highlights the influence of each group, measured by the total number of reposts originating from these opinion leaders. Notably, the influence trends align closely with the observed within-group connections. These findings suggest that the presence of echo chambers on China's social media, separating official and unofficial users, varies depending on the specific climate change event. Certain geopolitical events appear to disrupt these chambers, fostering increased mutual engagement and breaking traditional silos.

Main framing of climate change adopted by opinion leaders

Media framing is the process of organizing and presenting information, involving the selection of certain aspects of perceived reality to make them prominent. Our classification of the main frames on Weibo refers to existing climate narrative methods, which lay the foundation for comparing and enriching the public's perception of climate change reported in previous studies[34]. Common climate narrative frames include climate change opposition framework, global governance framework, climate justice framework, and ecological crisis framework[35–37]. Although these frameworks provide valuable perspectives, in-depth analysis of climate narratives needs to be further refined and supplemented based on actual data, especially in the social and cultural context of China. Therefore, we propose a climate narrative framework for Chinese social media by integrating insights from the above literature and our text analysis results. Through the word co-occurrence analysis, the top 135 phrases across opinion leaders' Weibo posts were classified into seven main types of climate change frames, and their connections are illustrated in Fig. 4A. Detailed phrases related to each can be found in Table S6.

The "China-US relations" revolves around phrases concerning US presidents, advocating for collaboration, and debating controversies related to COVID-19. The "personal actions" focuses on day-to-day behaviors that individuals can undertake to mitigate climate change. The "political will" frame emphasizes the commitment to address climate change, with typical phrases including carbon peak and ecological civilization. The "mitigation" is characterized by phrases highlighting emission reduction goals and measures for transitioning to the low-carbon economy. The "global initiative" includes keywords related to international negotiations and diplomatic concepts. The "scientific reality" focuses on scientific findings or reports concerning the causes and impacts of climate change, often containing imageries like polar bears. Lastly, the "economy and society" underscores how addressing climate change aligns with the pursuit of high-quality economic development.

Examining the five representative peak events as case studies, we investigated the popularity of frames among opinion leaders to examine Hypothesis #2 (Fig. 4B). We also directly compared the percentage of using different frames between official and unofficial user groups (Fig. 4C). On average, unofficial users are more likely to adopt frames such as China-U.S. relations, scientific realities, socio-economic impacts, and personal actions. In contrast, users with official backgrounds emphasize political will and mitigation policy frames. A chi-square test of association between user background and frame adoption reveals statistically significant differences in certain frames (Table S7), while other user indices are not the main factors affecting the adoption of frames (Table S8).

Central official media and government agency accounts tend to focus more on national political themes compared to other official accounts. During COP27, 55.8% of posts from official accounts employed the "global initiatives" and "political will" frames, significantly higher than the 21.0% from unofficial accounts. Similarly, during the 2022 "Two Sessions," 61.6% of posts from official accounts focused on frames like "political will" and "mitigation," compared to 33.6% from unofficial accounts.

In contrast, unofficial accounts, primarily personal, business, and some unofficial media outlets, engage more with frames related to scientific realities, socio-economic impacts, and China-U.S. relations. For instance, following Trump's 2017 announcement of the U.S. withdrawal from the Paris Agreement, 43.2% of posts discussing the event came from these groups, reflecting how climate discussions intersect with personal lives and international affairs. Additionally, 66.0% of their posts during the 2018 summer heatwave focused on highlighting the scientific realities of climate change and promoting personal actions. These findings underscore the diverse strategies used by different groups in shaping climate discussions on Chinese social media. Framing preferences are strongly influenced by account types and their primary audience demographics.

Discussions

In this study, we conducted an in-depth analysis of the discussions regarding climate change on China's social media platforms over the past decade. Our investigation uncovered notable patterns driven by periodic events and identified influential figures alongside their strategic framing within the context of Chinese culture and political system. We have summarized several unique landscapes in terms of main participants, opinion leaders, network structures, and the content of discussions about climate change between official and unofficial users. These findings highlight the importance of tailored strategies to promote global climate action at the local level and through digital new media platforms.

On Chinese social media platforms, grassroots personal accounts remain the key participants in discussions about climate change, constituting more than four-fifths of engaged accounts during peak events. Nevertheless, state-owned official media and certain local government agencies show higher levels of public engagement, especially regarding topics related to domestic climate change policies. This pattern can be explained through agenda-setting and source credibility theories. First, government campaigns and educational initiatives are typically organized in a top-down manner to promote local conservation and advance policies[38]. Second, official media outlets are perceived by the Chinese public as more trustworthy, authoritative, and thus more inclined to be engaged with[39,40]. In contrast, individual political figures, radio commentators, scientists, and journalists, who are reported as key actors in Western studies[41], have less impact in the Chinese context.

In examining social networks, we have observed unique patterns regarding user interaction and the exchange of opinions on social media

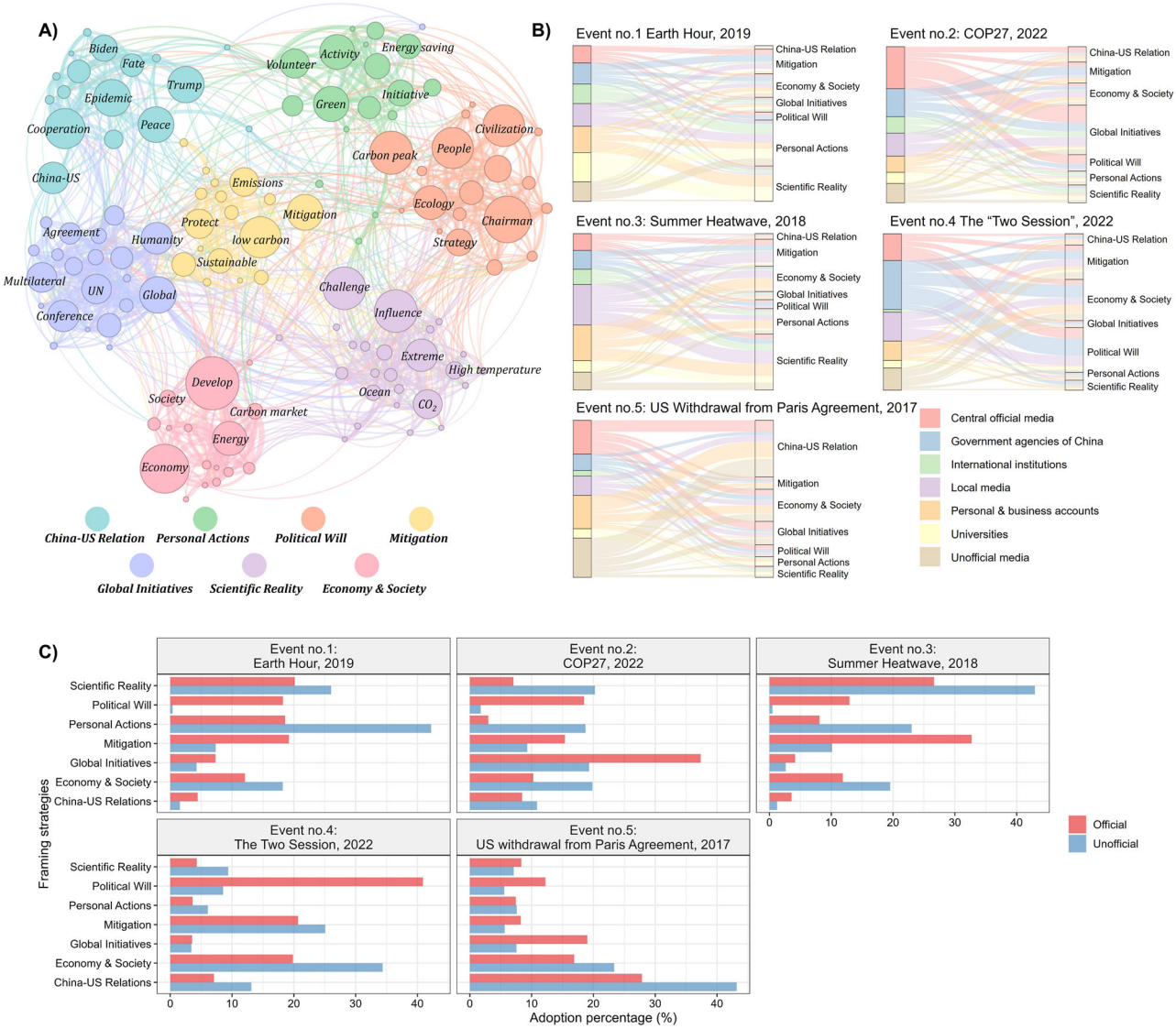

Fig. 4 | Framing of climate change and its utilization by opinion leaders on the online social media platform. A The word co-occurrence network illustrating the main phrases and their communities. Seven primary frames were identified and delineated using distinct colors. **B** Distribution of frame popularity among various types of opinion leaders during five representative peak events. **C** The use of climate narrative frames by official and unofficial users across five events.

platforms. Existing studies have shown that strong political polarization fosters fragmented climate change opinions and forms distinct information chambers[5,42]. Conservatives are more likely to express skepticism or denial of climate change, while liberals are inclined to acknowledge its existence and advocate for action[43,44]. This divide is evident in the content shared and discussions held on platforms like Twitter and Facebook, where users often segregate into two main networks and engage in debates that align with their ideological leanings[45]. In China, the scientific consensus on climate change is widely disseminated and accepted, and there have been numerous national surveys that show climate skepticism is not prevalent in the country[10,46]. In this study, we found that climate change-related networks in China are generally sparse and small, primarily driven by similar entities such as universities, unofficial media, and government agencies. Our findings partially support Hypothesis #1, indicating the presence of an echo chamber effect between official and unofficial users, but this effect varies depending on specific real-world events. For example, issues like China-U.S. relations appear to disrupt this echo chamber. Besides, different framing strategies are employed by these two groups to engage the public, disseminate scientific consensus, and interpret domestic climate policies, which supports Hypothesis #2.

Our findings indicate that the Chinese public generally holds a positive view of the country's efforts in addressing climate change. This sentiment extends beyond the promotion of domestic climate policies, which previous studies have suggested may be influenced by state-owned media and government agencies[16]. It is also evident in numerous online activities where personal accounts engage in discussions favoring green transition and sustainable development. However, there are three categories of real-world events, namely haze crises, COVID-19, and China-US relations, that tend to diminish positive response. Previous research highlights a growing awareness and concern about environmental issues, leading to a misconception of global climate change as local air pollution[10]. This may result in increased levels of concern and anxiety during haze events[47,48]. COVID-19, as a global crisis, has also affected the Chinese public emotions. We found a greater expression of worries and concerns, reflected in significantly lower sentiment scores. Finally, Weibo platform questions perceived inconsistency and reversals in U.S. climate policy, typically accompanied by negative emotions. It fosters a narrative that frames climate change not only as an environmental challenge but also as a strategic arena in international power dynamics[49].

Several strategies to promote climate change initiatives can be used within the Chinese context. While we do not find strong political

polarization regarding opinions on climate change, there are numerous smaller networks fragmented across groups. For instance, young people and females are more inclined to participate in online activities initiated by celebrities or universities. It is crucial to foster targeted communication initiatives and enhance interactions among a wider range of users. Then, our analysis revealed that extreme weather events triggering intense online responses were primarily limited to heatwaves or coldwaves. More localized hazards, such as floods, hurricanes, and wildfires, were not adequately detected in our study. It is vital to seize opportunities during natural hazards to reinforce the connections between climate change and catastrophic events, thereby enhancing public awareness.

Our data and methodology offer a valuable tool for tracking and understanding the Chinese public's responses to climate change issues. However, we acknowledge several inherent limitations in utilizing the Weibo big data collected in this study. First, our analysis is limited to understanding the communication patterns of Weibo users, which hampers our ability to explain phenomena in regions with limited development and among elderly populations due to their lower internet access and preference for social media usage. Second, our study focuses on analyzing influential accounts rather than capturing the voices of the general public. For example, Weibo discussions related to "extreme weather" events are mainly influenced by weather forecasts from news media, leading to relatively positive emotions compared to our initial expectations. Third, the data collected from social media are event-driven, providing insights into public responses to specific real-world issues. However, it remains unclear how individuals conceptualize and interpret these complex climate change issues themselves. Employing heuristic methods becomes necessary to gain insight into what climate change truly means to them[50]. Finally, the Weibo data are analyzed at a group level and lack detailed user information such as socio-demographics, personal beliefs, and cultural contexts. In addition, some information provided in user profiles cannot be verified for accuracy. Combining social media big data with traditional questionnaires is essential to resolve these challenges and enhance the validity and reliability of the findings.

Methods

Data collection and preprocessing

Sina Weibo is a massively popular micro-blogging social media site in China similar to Twitter (now 'X'). Users are limited to 2000 Chinese characters per post, with only the initial 140 characters visible at first glance (a link is provided to access the full post). Users may incorporate external links, images, multimedia content, and emoticons in weibos. We developed an algorithm using Scrapy, an open-source crawling framework in Python[51]. This algorithm automatically retrieved historical posts based on specific keywords through Weibo's application programming interface (API). The data mining process workflow was outlined in Text S1. In this study, we employed the Chinese term for "climate change" as the keyword to filter the search sequence during the summer of 2023. The historical search timeframe was set from January 2012 to December 2022. Data collection began in March 2023 and took 6 months to complete. The final dataset has a total of 5.3 million original posts from 2.2 million distinct users from 31 Chinese provinces.

The obtained data fields include the original text content of the Weibo posts, number of likes, comments, or reposts, and publishing time. We also retrieved user-related information, including user ID, account name, verification status, personal introduction, number of followers and followings, geographic location, etc. Weibo verification is the process of obtaining an official account status. The main verification types encompass personal accounts (celebrities, commenters, scholars, etc.) and organizations or institutions (enterprises, government agencies, news media, etc.). This facilitates our identification of key influencers. The location of each account is determined based on the user's most recent geo-tagged posts or self-reported geolocation in their profile (Fig. S11). The data preprocessing process is outlined in Text S2. We performed a correlation analysis between the number of users and the total population for each province. The results showed a significant correlation, affirming the spatial representativeness of our data (Fig. S12).

We then performed a few data preprocessing steps for further textual analysis. We first detected and removed robots, here defined as accounts posting more than 50 Weibo posts per day. Duplicate posts made by the same user on the same day were identified and removed, keeping only one post. After removing the unqualified data as described, this study obtained a final database of 3.1 million valid posts from 1.3 million users. We then cleaned redundant information from original texts such as stop-words, URLs, whitespaces, numbers, and punctuations. The preprocessing efforts ensured clean and relevant textual data for downstream mining and modeling tasks.

Major events detection

To detect real-world events that may trigger hot online debates of climate change in Chinese online social media platform, here we defined an Online Attention Metric (OAM) that measures the daily dynamics of interest a real-world event receives. The metric is defined as formula (1).

$$\mathrm{OAM_w} = \sum_{t=1}^{7} \mathrm{Engagement_t} \times \mathrm{Popularity_t} \tag{1}$$

The $\mathrm{OAM_w}$ was computed weekly, generating a time series reflecting public attention to climate change issues on the Weibo platform. This metric consists of two primary parameters. $\mathrm{Popularity_t}$ computes the total number of Weibo posts related to climate change per day. $\mathrm{Engagement_t}$ serves as a weighting variable, quantifying the level of public engagement with each post. The formula (2) below illustrates the calculation process for $\mathrm{Engagement_t}$:

$$\mathrm{Engagement} = \log_{10}(\mathrm{likes} + \mathrm{comments} + \mathrm{reposts}) \tag{2}$$

Three fields related to different types of engagement with each post were considered, including the number of likes, comments, and reposts. We computed the average sum of these counts, then applied a common logarithmic transformation to reduce the influence of extreme numbers, and the min-max normalization to convert each data value to a range between 0 and 1.

We then applied a robust peak detection algorithm based on Z-score on the weekly OAM streams, to detect hot-debated online events. The principle is that if the OAM for a week is a certain number of standard deviations away from a moving average, the algorithm signals a significant peaking event[52]. In this study, the Z-score of the OAM per week is calculated using the formula (3) below, with a lagging window of 5 weeks and a threshold set at 3 standard deviations away from the moving average to be labeled as a peak. A total of 48 spikes in online discussions about climate change in China were identified between 2012 and 2022.

$$z = \frac{\mathrm{OAM_w} - \mu_w}{\sigma_w}$$
$$\mu_w = \frac{1}{5}\sum_{i=-2}^{2} \mathrm{OAM_{w+i}} \quad \sigma_w = \text{standard deviation} \tag{3}$$

Topic modeling and sentiment analysis

To link each peak with a corresponding real-world event, we collected all Weibo posts published on those dates and then applied Biterm Topic Modeling, an advanced Natural language processing (NLP) technique to help extract valuable information from text data. Topic modeling is effective in identifying topics or themes present in a collection of documents. Biterm Topic Modeling (BTM) is one of the most cutting-edge algorithms, especially suited for short-text analysis. Recent literature has reported that BTM outperforms some traditional topic models, such as Latent Dirichlet Allocation (LDA), when dealing with short texts like tweets and Weibo posts[53]. We inputted posts from the selected peak date period into the BTM for topic modeling. We determined the optimal number of topic categories to be 5 based on the perplexity of the computational topic model (Fig. S13). The

topic model outputs the top few keywords for the 5 topics and the topic distribution probability θ for each post, these details and results are detailed in Text S3 and Table S9. We labeled each post with the highest θ as the topic of this post, which finally categorized them into five groups.

To measure the sentiment of climate change Weibo posts, we employed a Pre-trained Language Model, Chinese-BERT-wwm, released by the Joint Laboratory of HIT and iFLYTEK Research[54] for analysis. This is an enhanced version of the state-of-the-art big language models in text classification, i.e., Bidirectional Encoder Representations from Transformers (BERT). The BERT model is built upon Transformers, a deep learning architecture that can understand word in a sentence by considering both left and right contexts[55]. It is achieved through pre-training on extensive collections of text data. Chinese-BERT-wwm is specifically customized for processing Chinese short texts, having been pre-trained on a dataset of one million Chinese tweets labeled with positive or negative sentiment. It has demonstrated a remarkable accuracy of up to 95% in sentence sentiment classification[56]. In our study, we randomly sampled 1000 instances from our dataset and compared the classification performance of Chinese-BERT-wwm with traditional techniques. The results showed that the model achieved an accuracy improvement of 30% and 10% compared to dictionary matching approaches and neural network models like Long Short-Term Memory (LSTM) (Tables S10 and 11). Table S12 shows the robustness analysis of sentiment analysis and topic models. The limitations of sentiment analysis and topic models are stated in Text S4.

Social network analysis based on text similarity

To explore the process of information spread on social media platform and identify the role of influencers, it is essential to build a network that illustrates the engagement dynamics among users, such as reposts. However, a significant challenge with Weibo data collected from APIs is the absence of post-repost relationships between accounts. We addressed this challenge by introducing a network analysis technique based on text similarities. This approach overcomes the limitation of the absence of post-repost links between accounts and offers the advantage of identifying influencers who share similar opinions, even in the absence of direct interactions.

To assess text similarities among Weibo posts, we utilized a word embedding model based on BERT to convert each post into a continuous vector space. By calculating the cosine angle between vector representations, we measure the extent of similarity between two Weibo posts in relation to their word usage and frequency. More technical details for calculating text cosine similarity can be found in Text S5. We conducted text similarity analysis on Weibo posts related to the five event categories. The results indicated that during the early phase of each event, a large number of posts exhibited high similarity (Fig. S14). It presents that influential accounts first published posts, which were then copied or slightly modified by their followers. This similarity gradually decreased over time, suggesting a potential dissemination pattern for these climate-related posts and more diverse interpretations by other accounts. Hence, we can infer opinion similarities between two users and their potential post-repost connections.

Therefore, we chose the event with the highest peak from each event category and constructed the corresponding Weibo text similarity matrix for building networks. For each event category, we identified the top 100 most influential accounts as nodes. Source and target nodes were determined based on the time of posting. The influence of an account was quantified by the sum of likes, reposts, and comments received from its posts related to the specific event. Only connections between two nodes with a text similarity exceeding 0.7 were used as the edges. To validate the robustness of our findings, we calculated several centrality measures to identify key nodes in the network, including PageRank, eigenvector centrality, and betweenness centrality (Table S13). The correlations between these centrality measures were above 0.5 for all events, with some exceeding 0.8.

The network was then analyzed in open-access software Gephi[57]. To visualize the network, we chose a circular layout and the gravity was set as 0.8. A Fruchterman Reingold algorithm was used to detect the distribution pattern by detecting the main modularity and interactions among nodes[58]. It allows nodes with inherent connections to be closer, while nodes without connections tend to be repelled, thus forming distinguishable community structures within the network. The thickness of edges was determined by text similarity, and the node size was determined by their weibos' overall influence and public engagement. This approach allows us to gain a more insightful understanding of the propagation patterns and potential dependencies among users.

We proceeded to manually classify these opinion leaders based on account bio and description, verified account types, and primary content focus. These accounts were sorted into seven categories: central official media, local media, unofficial media, universities, government agencies of China, international institutions, and personal and business accounts. We then identified the top twenty most influential users for each event and indicated their account types on the circular network plot to analyze their roles as opinion leaders in climate change discussions. To test the two hypotheses proposed in this study, we further categorized these users into official and unofficial accounts.

Identifying framing of climate change used by opinion leaders

Analyzing the framing of climate change employed by opinion leaders is crucial due to the close connections of social media users, which can lead to the amplification of certain viewpoints. To identify the framing strategies used by key opinion leaders, we systematically examined Weibo posts by influential opinion leaders identified through social network analysis. We tokenized these posts and calculated the frequency of occurrence for each tokenized phrase, ultimately generating a collection of 135 distinct phrases after removing duplicates. Next, we constructed a document-word matrix to quantify the distribution probability of key terms in each document. By multiplying this matrix with its transpose, we derived a word co-occurrence matrix, where higher values indicate a greater likelihood of two words appearing together and thus reflecting similar frames.

Utilizing this word co-occurrence matrix, we created a network where phrases served as nodes and their co-occurrence frequencies as weighted edges. Employing a Louvain community detection algorithm, we categorized these phrases into seven distinct groups, with a modularity output of 0.529, indicating a reasonable classification (Fig. S15). More details about the process are presented in Text S6. We manually parsed the words involved in each group and identified them as seven main frames of climate change.

Finally, we sought to analyze how opinion leaders employed framing techniques to express their viewpoints by categorizing each Weibo post from opinion leaders into one of the seven framing categories. This process involved calculating the frequency of mentions for each of the 135 phrases in every Weibo post. We then assigned each post to the framing category that corresponded to the highest frequency of mentioned phrases within that category. We calculated the proportion of posts in which official and unofficial users employed specific frames during each event, providing a clear and comprehensive understanding of how framing varied among opinion leaders during these key moments. These calculations helped reveal their frame preferences.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The processed dataset can be obtained from: https://doi.org/10.5281/zenodo.14096539.

Code availability

The code can be found at the following link: https://github.com/njuYaxinhu/ChineseSocialMedia.

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Author contributions

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The authors declare no competing interests.

Additional information

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